

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
7th Semester of B. Pharm. Examination
University Theory Examination December 2015
PH405 DOSAGE FORM DESIGN

Date: 03.12.2015, Thursday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A** (i) Explain the following study parameters indicating their effect on formulation, stability & bioavailability: 05
(a) Particle density (b) Partition co-efficient
- (ii) Write a note on 'X-ray diffraction study' as a method to identify the polymorphism in the drug. 05
- B** Why melting point is studied in preformulation? Write a short note on role of DSC in preformulation studies with a schematic diagram of thermogram. 10
- C** Explain the importance of solubility in the preformulation studies. Illustrate your answer with suitable examples. Suggest different ways to manipulate the solubility of drug. 10

Q 2 Attempt any TWO of the following

- A** (i). Enlist ideal properties for 'Preservatives'. 05
(ii) Discuss in detail various factors affecting selection of an 'Anti-oxidant'. 05
- B** (i) Enumerate different classes of excipients for tablet dosage form. Discuss various factors affecting disintegration of tablets. 05
(ii) Write a detailed note on 'different classes of binders'. 05
- C** How do emulsifiers stabilize emulsion? Explain various emulsifiers in detail. 10

SECTION-II

Q 3 Attempt any **TWO** of the following

- A Define 'Dissolution of drug'. Explain in detail different theories of dissolution. 10
- B Explain in detail the principle and testing methodology of 'Accelerated stability study'. Write about the limitations of the study. 10
- C (i) Explain in detail the stability protocol of tablet dosage form. 06
- (ii) What do you mean by stability of pharmaceuticals? Write about the need and advantages of the stability testing. 04

Q 4 Attempt any **TWO** of the following

- A (i) Enlist various dissolution test apparatus as per USP. Explain the Type 1 apparatus and Type 5 apparatus in detail. 07
- (ii) Write about various applications indicating the necessity of in-vitro dissolution testing. 03
- B What are the aims and objectives of the ICH guidelines? Explain the structure of the committee and the four major categories of ICH guidelines in detail. 10
- C Explain the principles governing the chemical degradation of drug by hydrolysis and oxidation and write about the preventive measures of it. 10
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